

COMP526: Computational Methods for Scientists, Fall 2022

Assignment 3 - Due Wednesday 10/5/22

Under your home directory create a directory assign3. `cd assign3.`

Leave your work in file named calculus1D.c in directory assign3.

The purpose of this assignment is to calculate approximations for the derivative and the integral of a given function in the following collection of functions (all of them with argument of type double) without using pointers (use arrays with the maximum value for n and keep track of the actual value of n):

1. Natural logarithm
2. Tangent
3. Arc sine
4. Hyperbolic cosine
5. Hyperbolic arc tangent

You will need to write the following functions

1. You have to write a function that presents the index function and the name of the function per line for all the functions above and asks the user to select one function by its index (the function should check that the user selection is within the range of the number of functions). Print a one-line message confirming the user choice.
2. Then the program should ask for a valid interval $[a, b]$ of definition of the selected function (so, it should ask for two numbers of type double with $b > a$). Your code should check that $[a, b]$ are a valid domain for the function chosen. Print a one-line message confirming interval $[a, b]$.
3. The program should ask for a valid positive integer n , $n \leq 100$, that will be utilized to divide interval $[a, b]$ into n equal size subintervals. Check the integer is positive. Print a one-line message confirming the number of cells in which the interval $[a, b]$ will be divided.
4. For each interior point $x_j = a + j \frac{b-a}{n}$, $j = 1, \dots, n-1$, compute $dh(x) = 2\sqrt{\epsilon \left| \frac{f(x)}{f''(x)} \right|}$ where $\epsilon = 10^{-16}$. Note that you will need to define the functions for each of the second derivatives.
5. The way to approximate the derivatives at x_j , $j = 1, \dots, n-1$ is by

$$f'(x_j) \approx \frac{f(x_j + dH(x_j)) - f(x_j - dH(x_j))}{2dH(x_j)},$$

where $dH(x) = \min\{dh(x), \frac{b-a}{2n}\}$.

6. Compare the error of the first derivative approximation at each x_j , $j = 1, \dots, n-1$. Notice that you will need to define the functions for the first derivatives of each of the functions.
7. Use the Trapezoidal rule

$$\int_a^b f(x) dx \approx \frac{b-a}{n} \left(\frac{f(a)}{2} + \sum_{k=1}^{n-1} \left[f\left(a + k \frac{b-a}{n}\right) \right] + \frac{f(b)}{2} \right),$$

to compute an approximation of the integral of $f(x)$ on $[a, b]$.

8. Compute the error of the integral approximation. Note that you will need the functions for each of the antiderivatives and use the Fundamental Theorem of Calculus to compute the exact integral.
9. You can test your code but the final version of the main function should contain only code for selecting the function, entering a, b, n and return 0;. This will facilitate my testing of your work.